

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:25

Annexure A

Concept Mapping

Code of Conduct:

S. Nambesh (192524559) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

S. Nambesh
Signature of the student:

COMPARATIVE CONCEPT MAP: OS TYPES

3)

Jobs Submitted as
batches (cards, tape)
Processor is
Sequentially assigned

Execution
flow

Simple to design
No interaction
(efficient for some jobs)

Advantages

low CPU utilization
(to wait)
lack of user interaction
High turnaround time

Limitations

Single-system view
Transparency
(location, access)

Networked
Nodes

Scalability
Fault Tolerance

Goals

Resource sharing over
network
High availability

Advantages

Complex communication
Synchronization
Network failure points
High overhead

Limitations

1. BATCH OS
(e.g., IBM OS/360)

2. TIME-SHARING
OS
(e.g., MULTICS)

3. Distributed OS
(e.g., Arceda,
Mach)

4. Real-time OS
(RTOS)
(e.g., Vxworks, QNX)

Similarities and Differences

Comparative
OS Types

Are all types of
operating system.

Manage
systems

Manage System
Resource

Share
resources

Manage
user
resource

Both require
Specialized
design

often use low-level
synchronization

Goal

- fairness and
- interactive access

Key feature

- Context Switching
- Multi Programming

Advantages

- Quick response time
- Efficient resource use
- Multiple users

Limitations

- Complex resource
management
(scheduling, memory)
- Security/protection needs

Constraint

- Time Critical responses
- Deterministic behaviour

Types

- Hard RTOS, Soft RTOS
(Missed deadlines acceptable)

Advantages

- Predictable Execution
- Minimal latency

Limitations

- High context-switching
overhead
- Less support for
generic multitasking

Both be single
- user/task

4)

DYNAMIC EXECUTION FLOW CONCEPT MAP

